Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/02/2006
PI Name:	Dhir, Vijay Ph.D.		
Project Title:	Investigations of Mechanisms Associated with Nucleate Boiling under Microgravity Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Los Angeles	**State:**	CA
Zip Code:	90095-8357	**Congressional District:**	33
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	05/01/1996	**End Date:**	05/15/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2467		
Performance Goal No.:	1.23		
Performance Goal Text:	Pursue ground-breaking basic research in fluid physics and transport phenomena to provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.		

Task Description:	Task 1: Fabrication of the Experimental Setup. Under this task, the test section and liquid holding and viewing chambers will be fabricated. Artificial cylindrical cavities will be formed on aluminum wafers. A single cavity and two or four cavities, with a prescribed spacing and size, will be formed. The desired nucleation wall superheat will be used to determine the size of the mouth of the cavities. Task 2: Experiments. The basic experiments for flow and temperature field around single and multiple (2 or 4 separated or merged bubbles growing on downward facing or inclined surfaces) will be carried out under normal gravity conditions. The experiments will be conducted at one atmosphere pressure, but liquid subcooling will be varied from 0 to 10°C. Water and PF-5060 will be used as test liquids. Task 3: Analytical/Numerical Models. In this task, transient temperature and flow field in vapor and liquid will be determined during growth of a single bubble. Analysis will include the evolution of the vapor-liquid interface and development of microlayer underneath the bubbles. For merged bubbles, detailed calculations of flow and temperature field will be carried out for transient shapes of vapor stems supporting a large bubble and the corresponding evaporation rate. Flow and temperature field for a bubble sliding along a heated wall will also be determined. Microgravity conditions will be simulated and a framework of a numerical tool for prediction of nucleate boiling heat fluxes under microgravity conditions will be developed. Task 4: Experiments in a KC-135. To understand bubble growth and detachment behavior of single or large merged bubbles, boiling experiments will be conducted under low gravity (10-2 g) conditions of the aircraft. In these experiments, designed surfaces will be used. Visual observations and heat transfer data will be taken, but holography will not be used. The apparatus used for laboratory experiments will also be employed for experiments in the aircraft. Task 5: Experiments in the Space Station. Effort will be devoted for defining a boiling experiment to be conducted on a designed surface. The experiment will provide microgravity data on bubble growth and departure. These data are needed for development of a credible model for nucleate boiling heat fluxes under microgravity conditions. The heat transfer data will also be obtained and will be used to validate the models. In this work a building block type of approach is used so that a basic understanding of the processes that contribute to nucleate boiling heat fluxes under microgravity conditions can be developed. This understanding will lead to development of a mechanistic model for nucleate boiling heat transfer which could eventually be used as a design tool in space applications. Boiling is known to be a very efficient mode of heat transfer, and as such, it is employed in component cooling and in various energy conversion systems. For space applications boiling is the heat transfer mode of choice, since for a given power rating the size of a component can be significanlty reduced. Applications of boiling can be found in the areas of thermal management, fluid handling and control, power systems, on-orbit storage and supply systems for cryogenic propellants and life support fluids, cooling of electronic packages for power systems associated with various instrumentation and control systems, and various processes associated with ISRU.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Task 1: Complete. Task 2: Complete. Task 3: Completed, Continuing for more than 5 bubbles and commercial surfaces. Task 4: Complete. Task 5: Continuing, currently in planning and development.
Bibliography Type:	Description: (Last Updated: 08/23/2012)
Articles in Peer-reviewed Journals	Mukherjee A, Dhir VK. "Study of lateral merger of vapor bubbles during nucleate pool boiling," Journal of Heat Transfer. 2004 Dec;126(6):1023 (17 p.). http://dx.doi.org/10.1115/1.1834614 , Dec-2004
Articles in Peer-reviewed Journals	Dhir VK, Abarajith HS, Warrier G, Son G. "Bubble Dynamics and Nucleate Boiling Heat Transfer- Numerical Simulation and Experimental Validation." Journal of Energy Heat and Mass Transfer. In press as of March 2005. , Mar-2005
Papers from Meeting Proceedings	Abarajith HS, Dhir VK, Son G. "Numerical Simulation of the Dynamics of Multiple Bubble Merger during Pool Boiling under Reduced Gravity Conditions." Japan-US Seminar on Multiphase Flows, Japan, December 6-11, 2004. Proceedings of Japan- USA Seminar on Multiphase Flows, Japan, December 6-11, 2004. , Dec-2004
Papers from Meeting Proceedings	Dhir VK, Abarajith HS, Warrier GR. "From Nano to Micro to Macro Scales in Boiling." NATO Advanced Study Institute on Microscale Heat Transfer - Fundamentals and Applications in Biological and Micromechanical Systems, Cesme-Izmir, Turkey, July, 2004. In: Microscale heat transfer: fundamentals and applications. Proceedings of the NATO Advanced Study Institute on Microscale Heat Transfer - Fundamentals and Applications in Biological and Micromechanical Systems. Edited by S. Kakac et al. Dordrecht, The Netherlands : Springer, 2005. p. 197-216. (NATO science series. Series II, Mathematics, physics, and chemistry ; v. 193) , Jun-2005